

Component	Interface	Estimated Current (mA)	Estimated Power (W @ 5V)	Estimated Weight (g)
Arduino Uno	Main microcontroller	45	0.225	25
Internal Temp Sensor	Environmental sensing	1.5	0.0075	2
External Temp Sensor	Environmental sensing	1.5	0.0075	2
Pressure/Humidity Sensor	Atmospheric sensing	1.0	0.005	3
Accelerometer	Motion sensing	4	0.020	2
MicroSD Module	Data storage	50	0.250	5
LEDs	System status	20	0.100	1
Battery pack	Power source	N/A	N/A	60
Wiring	Connections	N/A	N/A	15

Total Estimated Current = ~130 mA

Total Estimated Power = ~0.62W

Total Estimated Weight = 115g

Feasibility of Existing Design

The estimated power consumption is approximately 130mA, which corresponds to 0.6W of power at a 5V input voltage. The estimated mass of the whole system is also around 115g. These values indicate that this system is feasible to construct, as the power requirements are relatively low and the total mass remains below the 200g constraint.

However, since some components in the original kit are not available, the existing design can't be reproduced using the same parts. Certain parts may need to be replaced with alternatives, specifically some sensors - perhaps a single sensor may be used to combine functionality from several individual sensors.